Overview

The hardware consists of 2 motors, pan and tilt, 2 L pieces, top and bottom, and a camera. The pan motor at the bottom attaches to the bottom L piece which attaches to the tilt motor. The tilt motor attaches to the top L piece which holds the camera.

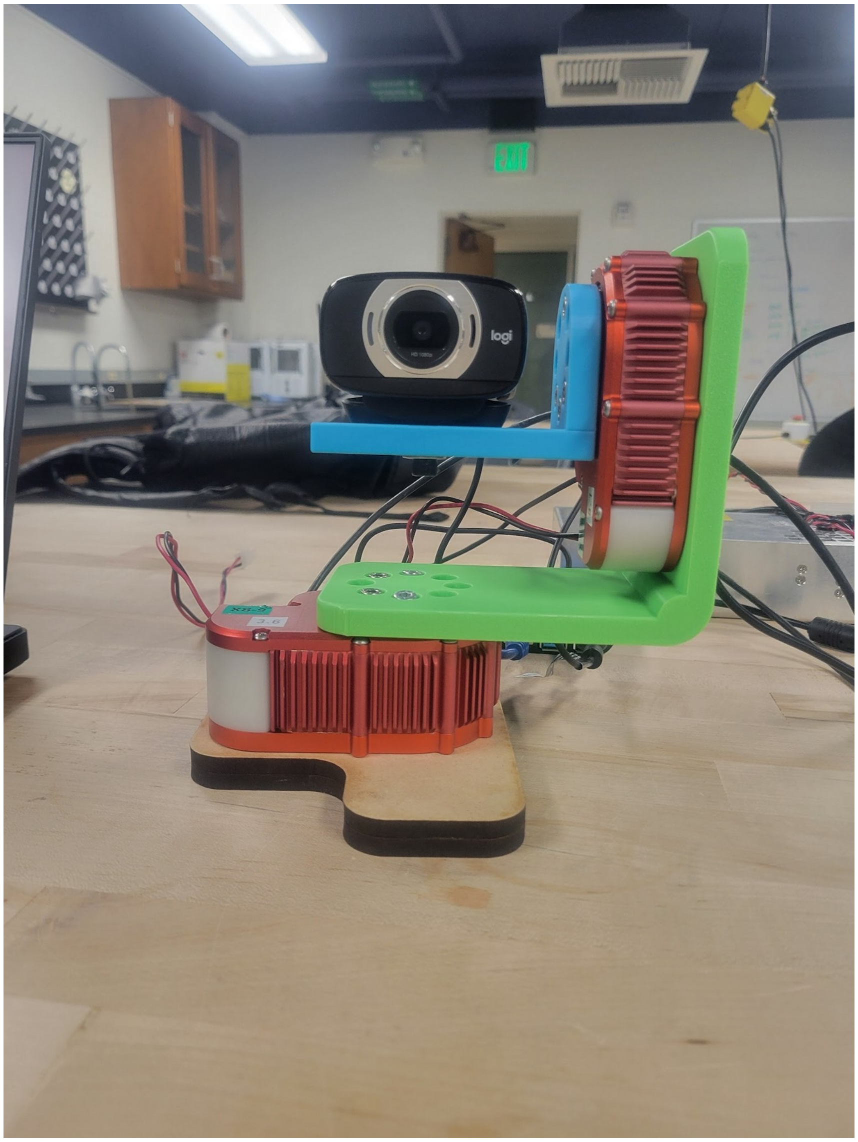
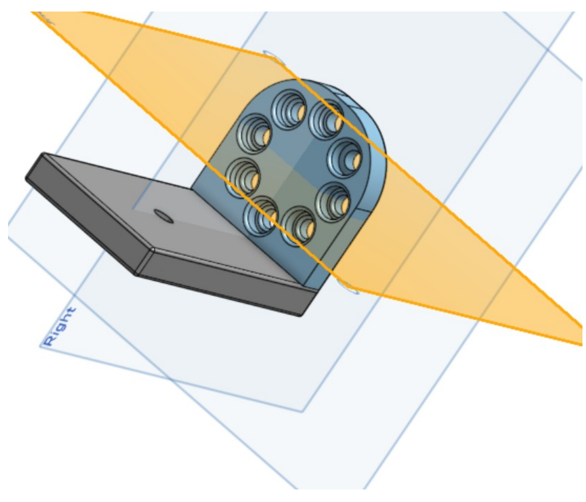


Fig. 1 *depicts the final assembled hardware.*

Top Piece:

This piece is colored cyan and controls the tilt of the camera. It has the circular screw pattern on the rounded piece to attach to the tilt motor, and a singular screw hole on the sharper piece for the camera. The expected **range of motion** is **120°** in either direction.



Bottom Piece:

This piece is colored lime and controls the pan of the camera. This piece has the same circular screw pattern to attach to the top of the pan motor. The other face has a partial square screw pattern in order to attach to the back of the tilt motor. The expected **range of motion** is **120°** in either direction.

Fig. 2 *depicts the final CAD design of the top L piece*

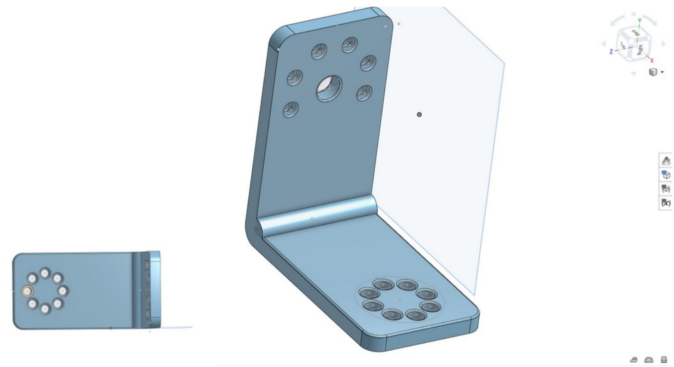


Fig. 3 *depicts the final CAD design of the bottom L piece*

Detailed CAD Drawings

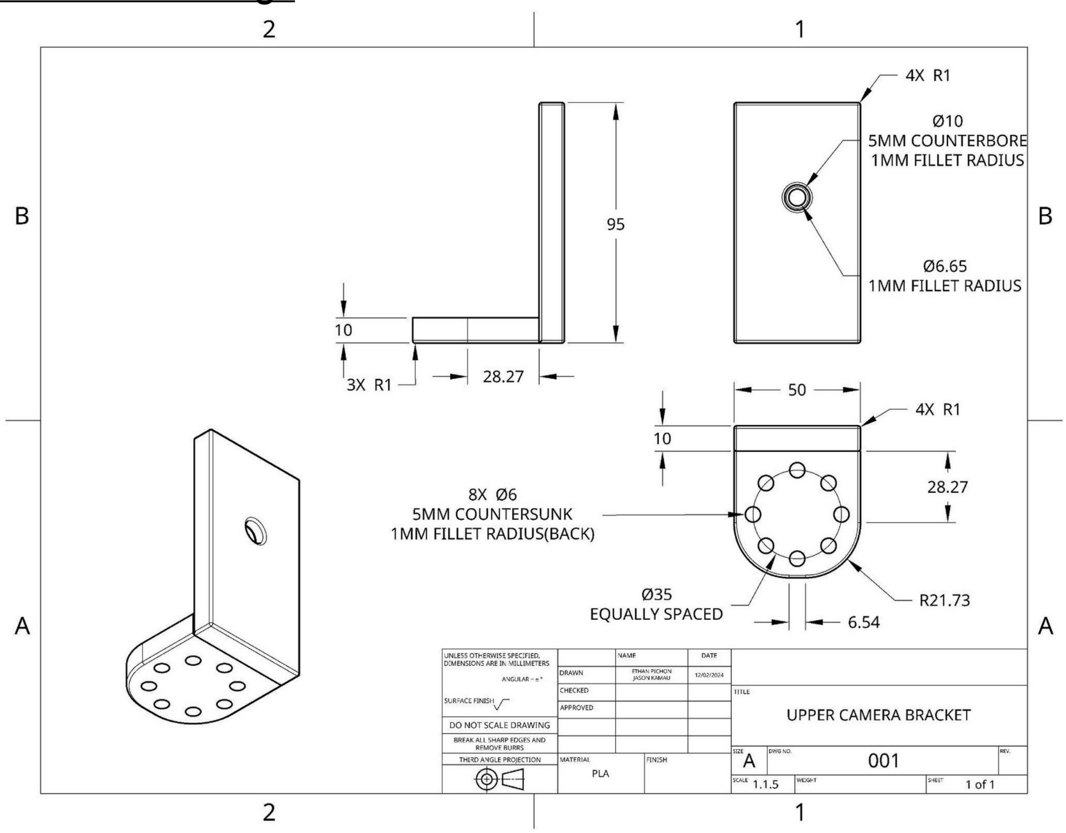


Fig. 4 *depicts the detailed CAD drawing of the Top L bracket including the dimensions and material used for the piece.*

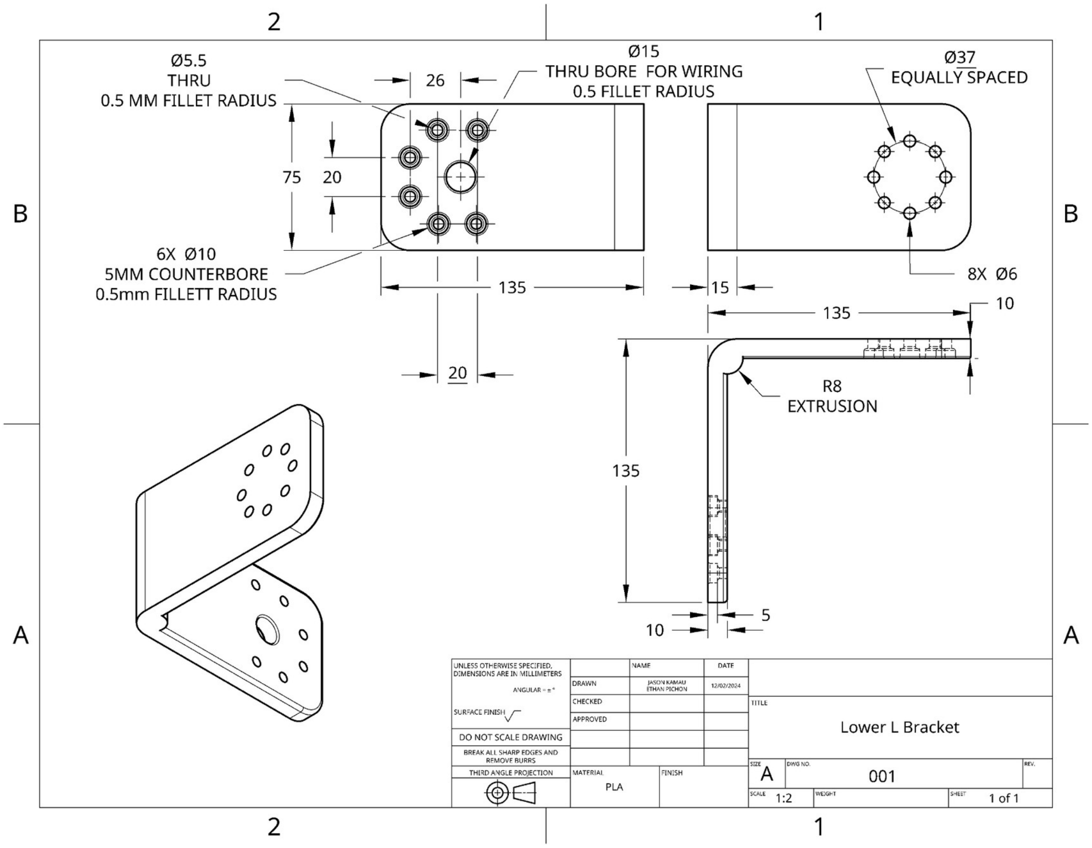


Fig. 5 *depicts the detailed CAD drawing of the Bottom L bracket including the dimensions and material used for the piece.*

Evaluation of the Print

Table 1

Printing Parameter	Top Piece	Bottom Piece
Color	Cyan	Lime
Material	PLA	PLA
Style	Strengthened(thicker walls)	Strengthened(thicker walls)
Supports	Yes(Tree Supports)	No

Table 1 *shows the printing parameters used for both the L pieces.*

Evaluation:

For the Top piece, the circular hole pattern was specified to be 10 mm for the outer circle and 6 mm for the inner circle. The actual print came out to be 0.1 mm smaller than specified. The singular hole for the camera had a similar error. The tree supports also created ridges in the holes which further lowered the effective hole size. This is why the additional millimeter of tolerance was necessary for the printing. The piece was printed with the circular screw pattern on the bottom. This reduced the number of supports necessary for the screw holes and thus reduced the ridges in those holes compared to the camera hole.

For the bottom piece, the circular hole pattern was specified to be 10 mm for the outer circle and 6 mm for the inner circle. The actual print came out to be 0.1-0.5 mm smaller than specified. The partial square hole pattern was specified to be 10 mm for the outer circle and 5.5 mm for the inner circle. The actual print came out to be 0.1-0.5 mm smaller than specified.This increased error is likely due to the lack of supports and that the piece was printed on its side so all the holes were printed sideways. This made the 1 mm of tolerance even more necessary for this piece.

Self-Evaluation/Improvements

Top Piece:

This piece was constructed with minimalism to reduce the motors' weight strain. This is why the piece is rounded at the top to cut down on filament. If we were to remake this piece, we could further reduce the size of the L by trimming the sides of the extrusion that holds the camera. All screw holes and rough edges were additionally filetted by 1 mm to give the screws more room, but also to reduce the chance of injury. All screw holes were also sunk in by 5 mm, so the screws could reach the motor/camera holes. If we were to redo the piece, we would give the camera hole a smaller counter sink as the screw didn’t fit well and needed to be secured with a stopper. Additionally, we would place the camera hole further away from the wall to fit the larger camera.

Bottom Piece:

This piece was designed to support the weight of the two motors, the upper bracket, and the camera hence the circular support design between the L shape. The upper design made space for the upper motor to face downwards to keep the center of gravity low hence increasing stability. Different shapes can be considered to reduce the amount of material and space used for support. The only part to reconsider in rebuilding is altering the diameter of the holes attached to the lower motor. Since the filament from the 3D printer is plastic, the process reduced the intended diameter by .5mm, and the screws had a hard time interlocking with the motor below.